SocioCore: A Four-Primitive Deterministic Social Flow Organism for Interpersonal Energy, Identity Coherence, Relational Stability, and Communication Load Governance

Social Energy Flow · Identity Coherence Flow · Relational Stability Flow · Communication Load Flow — The Lume 4/42 Architecture Applied to the Social Domain

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Companion Papers: BioCore · NeuroCore
Related: Lume 10.5281/zenodo.19382282 · Trust Layer 10.5281/zenodo.19560674 · DAIGS 10.5281/zenodo.19491784
Lume-V 10.5281/zenodo.19645097 · Lume-X 10.5281/zenodo.19443968 · Dissolution 10.5281/zenodo.15065493

Patent Pending — DarkWave Studios LLC
Systems and Methods for Deterministic Multi-Layer Wireless Energy Routing

Lume-X: Provisionally patented deterministic control runtime
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Abstract

I present SocioCore, a Lume-native deterministic social flow organism for interpersonal energy governance, identity coherence maintenance, relational stability monitoring, and communication load management. SocioCore implements the Lume 4/42 organism architecture — four irreducible flow primitives governing forty-two operational nodes — in the social domain. The four primitives are Social Energy Flow, Identity Coherence Flow, Relational Stability Flow, and Communication Load Flow.

SocioCore is the first organism in the Lume-Social vertical and occupies the social layer directly above NeuroCore in the canonical organism stack. Where NeuroCore governs the cognitive output of the individual, SocioCore governs the space between people: the social fields that emerge from and reciprocally shape the individual's biological and cognitive state. The two organisms are formally coupled: SocioCore receives NeuroCore and BioCore outputs as first-class input nodes and exports social state signals back to NeuroCore's Emotional and Cognitive Load primitives.

Unlike social wellness apps that provide generic recommendations, SocioCore formalizes social energy as a continuous flow field with threshold bands, routing rules, and hard constraint enforcement. The organism enforces five operating modes (Social Recovery, Connection, Boundary, Identity Stabilization, Communication Safety) selected deterministically by node pattern.

The paper includes a complete formal 2D/3D visual geometry specification with social pressure inward-gravity visualization, identity fragmentation crack effects, and both time-stack and depth-stack 3D modes.

Keywords: deterministic social governance, Lume synthetic organism, social flow, social energy, identity coherence, relational stability, communication load, social burnout, 4/42 organism architecture, interpersonal governance, Lume-Social vertical, NeuroCore integration, BioCore integration

1 Introduction

1.1 The Problem with Social Governance Today

Social neuroscience has established that social pain activates the same neural circuits as physical pain. Social isolation is classified as a public health crisis with mortality consequences comparable to smoking. Social belonging is a foundational human need whose absence produces measurable cognitive, immunological, and cardiovascular consequences. Yet no governance framework exists for social flow.

The gap is structural. Social governance has been treated as a soft, qualitative, unmeasurable domain. SocioCore challenges this assumption. Social energy is a continuous flow field. Identity coherence is a measurable state. Relational stability has detectable signals. Communication load has a threshold. All four can be governed.

1.2 The Social Flow Reframe

The foundational insight of this work is that social life is not a set of events. It is a continuous field of flows. Social energy flows between people in interactions. Identity flows across contexts — stabilizing or fragmenting depending on self-consistency under social pressure. Relationships hold states that evolve continuously. Communication accumulates as load that compounds when it exceeds processing capacity. SocioCore applies the Lume 4/42 organism architecture to this social flow system.

1.3 Position in the Lume Organism Stack

```
SOCIAL LAYER:     SocioCore (this paper)
COGNITIVE LAYER:  NeuroCore
BIOLOGICAL LAYER: BioCore
PHYSICAL LAYER:   Meridian / Verdara Ultra / HydroCore
```

SocioCore is the first Lume organism to govern a domain not localized to a single individual. Social flow exists in the space between people. A practitioner's SocioCore state is partially constituted by the states of the people around them.

1.4 Novel Contributions

1. The first formal 4/42 social flow organism specification
2. Social energy as a governable continuous flow with hard constraint enforcement

3. Identity coherence as a primary governance primitive with masking load, drift, and boundary monitoring
4. Five-mode deterministic social operating system
5. Critical pattern taxonomy (identity fragmentation risk: I4+R2+C2; social burnout risk: S3+C1+R6)
6. Tri-layer formal integration model (SocioCore↔NeuroCore↔BioCore)
7. Social pressure and identity fragmentation visual geometry

1.5 Scope and Limitations

SocioCore does not govern other people. It governs the practitioner's internal social state. Identity Coherence nodes I3 (Masking Load) and I4 (Identity Drift) have special relevance for neurodivergent practitioners and members of stigmatized groups. When Relational Stability nodes cross clinical thresholds, explicit professional referral is recommended.

2 Background and Related Work

2.1 Social Neuroscience

Eisenberger's discovery that social rejection activates the anterior cingulate cortex established the biological reality of social pain. Coan's social baseline theory proposes that the brain's default expectation is social proximity — social isolation is a metabolic overload event. SocioCore's Relational Stability primitive is grounded in this literature.

2.2 Identity and Self-Concept Research

Social identity theory, self-categorization theory, and the self-consistency motive research program collectively establish that identity coherence is a fundamental psychological need. Identity disruption produces anxiety, reduces self-regulatory capacity, and impairs decision quality. Goffman's dramaturgical framework formalizes the concept of role-switching and masking that SocioCore's I2 and I3 nodes capture.

2.3 Interpersonal Research

Gottman's longitudinal research identified specific markers of relational health. Attachment theory grounds R5 and R4. The emotional contagion literature establishes the mechanism by which others' emotional states directly influence the practitioner's own —

grounding R3 as a governance node.

2.4 Communication Load

Digital communication research has established that notification-driven messaging imposes significant cognitive and social costs. SocioCore's Communication Load primitive formalizes these costs as a continuous governance variable.

3 System Overview

3.1 Core Thesis

SocioCore ingests social signals, maintains social equilibrium, and outputs deterministic guidance. Equilibrium is not maximum social engagement — it governs for sustainable, coherent, safe social function.

3.2 The 4/42 Structure

```
Primitive State Vector SC:
    SC = [SocialEnergy, IdentityCoherence, RelationalStability,
CommunicationLoad]

Node State Ring R:
    R = [S1..S10, I1..I10, R1..R11, C1..C11]
    Each element is a normalized scalar in [-1.0, +1.0]
```

3.3 Temporal Scale Structure

- **Interaction-level:** State during or immediately after a social interaction.
- **Day-level:** Aggregate social state across a full day.
- **Week-level:** Trajectory of day-level states over seven days.

4 The Four SocioCore Primitives

4.1 Social Energy Flow

Governs capacity for social engagement, presence, and connection. Governed by ten nodes (S1–S10). S8 (Social Burnout Risk) is the primary hard constraint trigger.

4.2 Identity Coherence Flow

Governs stability and authenticity of self-experience across social contexts. Governed by ten nodes (I1–I10). I3 (Masking Load) captures the cost of sustained social performance departing from authentic self-expression. I10 (Identity Safety Index) is the composite terminal node.

4.3 Relational Stability Flow

Governs quality, safety, and predictability of relationships. Governed by eleven nodes (R1–R11). R1 (Trust Level), R5 (Attachment Security), and R11 (Relational Safety Index) are most foundational. R8 (Loneliness Load) has substantial mortality risk implications.

4.4 Communication Load Flow

Governs pressure, volume, and emotional weight of communication obligations. Governed by eleven nodes (C1–C11). C11 (Global Social Resilience) is the final integrative safety indicator.

5 The 42-Node Social State Ring

5.1 Social Energy Nodes (S1–S10)

Node	Name	Description
S1	Social Battery Level	Current capacity for social engagement.
S2	Interaction Fatigue	Cumulative depletion from social interactions over time.
S3	Social Overload	Social demands actively exceeding capacity.
S4	Social Isolation Pressure	Unwanted social distance driven by depletion or circumstances.
S5	Connection Warmth	Quality of positive relational feeling in current interactions.
S6	Presence/Attunement	Capacity for genuine presence in social interactions.
S7	Social Recovery Speed	Speed of social energy restoration after depletion.
S8	Social Burnout Risk	Probability of systemic social exhaustion. Primary hard constraint trigger.
S9	Social Curiosity/Openness	Orientation toward new social encounters. Leading indicator of withdrawal.
S10	Social Energy Stability	Consistency of social energy levels over time.

5.2 Identity Coherence Nodes (I1–I10)

Node	Name	Description
I1	Self-Consistency Across Contexts	Experience of being the same person across settings.
I2	Role Stability	Clarity and consistency of occupied social roles.
I3	Masking Load	Cost of presenting a social self departing from internal experience.
I4	Identity Drift	Progressive departure from core values under social pressure.
I5	Self-Trust	Confidence in own perceptions and judgments in social contexts.
I6	Boundary Strength	Capacity to maintain and enforce social boundaries.
I7	Authenticity Pressure	Pressure to present as different from authentic self.
I8	Identity Recovery Speed	Speed of restoring identity coherence after disruption.
I9	Narrative Coherence	Whether social life forms a coherent, self-authored story.
I10	Identity Safety Index	Composite terminal node for identity stability and protection.

5.3 Relational Stability Nodes (R1–R11)

Node	Name	Description
R1	Trust Level	Current trust in primary relationships. Prerequisite for relational safety.
R2	Conflict Pressure	Unresolved conflict level in relational environment.
R3	Emotional Contagion Sensitivity	Degree of influence from others' emotional states.
R4	Relational Predictability	Consistency and reliability of significant others' behavior.
R5	Attachment Security	State of attachment in significant relationships.
R6	Relational Overwhelm	Relational demands exceeding relational capacity.
R7	Social Support Strength	Access to adequate social support.
R8	Loneliness Load	Subjective experience of inadequate social connection. Major health risk.
R9	Relational Recovery Speed	Speed of restoring relational stability after disruption.
R10	Reciprocity Balance	Fairness in giving/receiving across significant relationships.
R11	Relational Safety Index	Composite terminal node for relational environment safety.

5.4 Communication Load Nodes (C1–C11)

Node	Name	Description
C1	Messaging Pressure	Volume and urgency of digital/analog communication demands.
C2	Emotional Labor Load	Cost of managing emotional displays in social/professional contexts.
C3	Miscommunication Risk	Probability of misunderstanding or failed meaning-transfer.
C4	Social Obligations Load	Weight of social commitments, duties, and expectations.
C5	Group Dynamics Stress	Stress from participation in group social contexts.
C6	Social Role Conflict	Stress from incompatible simultaneous social roles.
C7	Communication Clarity	Current quality and accuracy of communication.
C8	Social Bandwidth	Remaining capacity for meaningful social engagement.
C9	Conversation Fatigue	Cumulative depletion from sustained conversational engagement.
C10	Communication Recovery Speed	Speed of restoring communication capacity after depletion.
C11	Global Social Resilience	Integrated capacity to absorb additional social demands. Terminal safety node.

6 Node Threshold Architecture

6.1 Three-Band Model

- **Advisory:** Heightened awareness.
- **Caution:** Behavioral modification recommended.
- **Critical:** Hard constraint enforcement.

6.2 Personal Baseline Priority

Social state nodes are highly individual — introversion, neurodivergence, cultural context, and life stage all substantially determine optimal states. Minimum 14 days of self-report for calibration.

6.3 Normalized State Mapping

```
optimal: s = +0.3 to +1.0  |  advisory: s = 0.0 to -0.3
caution: s = -0.4 to -0.7  |  critical: s = -0.8 to -1.0
```

6.4 Critical Pattern Recognition

> **Identity Fragmentation Risk:** I4 + R2 + C2 all at caution → virtual critical, Identity Stabilization Mode evaluation.
> **Social Burnout Risk:** S3 + C1 + R6 all at caution → Social Recovery Mode, suppress expansion.

7 The Deterministic Social Engine

7.1 Inputs and Outputs

Inputs: Self-report; behavioral signals; social event log; NeuroCore outputs; BioCore outputs; context.

Outputs: Social mode selection; prioritized guidance; drift warnings; identity stabilization; relational repair guidance; communication load reduction; clinical referral when warranted.

7.2 The Six Social Engine Rules

Rule 1 — Hard Constraint: If any node crosses critical ($s \leq -0.8$) or critical pattern detected, suppress all social expansion recommendations and activate mandatory recovery/stabilization guidance.

> **Special triggers:** S8 at caution → all expansion suppressed. I4+R2+C2 pattern → Identity Stabilization Mode. R8 at critical → clinical referral alongside connection guidance.

Rule 2 — Social Energy Routing: When S2/S3 at caution: recommend social rest, reduce commitments, prioritize lowest-cost contacts. When S4 elevated with capacity: recommend gentle reconnection.

Rule 3 — Identity Routing: When I3/I4 at caution: reduce context switching, recommend journaling, increase safe-expression contexts. When I6 at caution: boundary reinforcement is primary guidance.

Rule 4 — Relational Routing: When R2 at caution: de-escalation guidance, defer high-stakes conversations when NeuroCore E3/E5 elevated. When R7 at caution: active support-seeking guidance.

Rule 5 — Communication Load Routing: When C1/C2/C4 at caution: notification silence, obligation audit, communication batching. When C9 at caution: silence windows, reduce meeting density.

Rule 6 — Trend-Aware Escalation:

Class	Pattern	Response
Spike	1–2 day degradation	Advisory adjustment only
Pattern	3–7 day sustained	Caution-level + active intervention
Collapse	7+ days or S10/C10 strongly negative	Critical intervention + clinical referral evaluation

8 Operating Modes

Mode	Trigger	Guidance
Social Recovery	S8 at caution; OR S2+S3 both at caution; OR burnout risk pattern; OR C11 at caution	Reduce discretionary commitments, increase solitude, lower communication volume. Expansion suppressed.
Connection	SC advisory+ across all primitives; S1 optimal; R7 adequate; C1 manageable	Invest in high-quality engagement. Prioritize depth over breadth. Highest-function mode.
Boundary	I6 at caution; OR C4 at caution; OR I3+I4 both at advisory	Identify active boundary erosions, practice declining language, reduce masking contexts.
Identity Stabilization	I4 at caution; OR I10 at caution; OR fragmentation pattern active	Reduce context switching, increase safe-expression time, values clarification. Clinical referral when I4/I10 critical.
Communication Safety	C11 at caution/critical; OR 3+ Communication Load nodes at caution	Complete communication triage. Minimum viable communication only. Maintained until C11 returns to advisory.

9 Integration Model — BioCore and NeuroCore

9.1 Three-Layer Coupling

SocioCore is formally coupled to both BioCore and NeuroCore. The coupling runs bidirectionally at each interface, forming a three-layer integrated governance stack.

9.2 BioCore → SocioCore

BioCore Node	SocioCore Node	Mechanism
M8 Inflammation	S2 Interaction Fatigue	Inflammation elevates irritability and social fatigue
S1 Acute Stress	C1 Messaging Pressure	Acute stress amplifies perceived urgency
S2 Chronic Stress	C2 Emotional Labor	Chronic stress reduces emotional labor capacity
N1 Sleep Duration	S1 Social Battery	Sleep deficit depletes social battery
N2 Sleep Quality	S7 Social Recovery	Poor sleep slows social recovery
S6 Burnout Risk	S8 Social Burnout Risk	Biological burnout is upstream driver

9.3 NeuroCore → SocioCore

NeuroCore Node	SocioCore Node	Mechanism
E6 Emotional Drift	R2 Conflict Pressure	Unexplained drift increases friction
L2 Chronic Cognitive Load	C1 Messaging Pressure	High load makes communication burdensome
E3 Anxiety Load	I4 Identity Drift	Anxiety destabilizes self-concept
L4 Overwhelm Index	C4 Social Obligations	Cognitive overwhelm amplifies obligation weight
E5 Irritability	R2 Conflict Pressure	Irritability increases conflict probability
I5 Decision Paralysis	I6 Boundary Strength	Decision paralysis erodes boundary maintenance

9.4 SocioCore → NeuroCore

SocioCore Node	NeuroCore Node	Mechanism
S3 Social Overload	L4 Overwhelm Index	Social overload contributes to cognitive overwhelm
R2 Conflict Pressure	E2 Emotional Reactivity	Active conflict elevates reactivity
R2 Conflict Pressure	E3 Anxiety Load	Conflict is a primary anxiety driver
I4 Identity Drift	I6 Narrative Coherence	Identity drift disrupts internal narrative
C2 Emotional Labor	L2 Chronic Cognitive Load	Emotional labor is chronic cognitive load
R8 Loneliness	E4 Depression Load	Loneliness contributes to depression

10 Visual Geometry

10.1 Social Color Palette

```
Social Energy Flow:         Warm coral/amber (#FF7043)
Identity Coherence Flow:    Sage green (#7CB87C)
Relational Stability Flow:  Soft indigo (#5C6BC0)
Communication Load Flow:    Warm grey/charcoal (#78909C)
```

10.2 The 4/42 Core and Ring

```
Core (four axes, clockwise from top):
  Up (0°):    Social Energy       — warm coral/amber
  Right (90°): Identity Coherence  — sage green
  Down (180°): Relational Stability — soft indigo
  Left (270°): Communication Load   — warm grey/charcoal
Ring (42 nodes):
  Arc 1: S1–S10   (coral)
  Arc 2: I1–I10   (sage)
  Arc 3: R1–R11   (indigo)
  Arc 4: C1–C11   (charcoal)
```

10.3 Cross-Primitive Links

- S1 (Social Battery) ↔ I6 (Boundary Strength)
- R2 (Conflict Pressure) ↔ C2 (Emotional Labor)
- I4 (Identity Drift) ↔ R1 (Trust Level)
- R8 (Loneliness) ↔ S4 (Isolation Pressure)
- C11 (Global Resilience) ↔ I10 (Identity Safety)

10.4 Domain-Specific Visual Elements

Social pressure: When Communication Load aggregate is at caution or below, a darkening inward-gravity halo wraps the left quadrant. **Identity fragmentation:** When I4 or I10 is at caution or below, hairline irregularities appear in the sage-green Identity axis and lobe. **Relational wobble:** When R11 is at caution or below, the indigo lobe gains a slow oscillation. **Loneliness signal:** When R8 crosses caution, the indigo lobe's inner edge dims.

10.5 3D Representation

- **Time Stack:** Z-axis encodes time. Node trajectories become vertical ribbons in primitive colors.
- **Depth Stack:** Three planes — top (conscious social behavior), middle (habitual patterns), bottom (implicit relational load from BioCore/NeuroCore inputs).

11 Global Invariant Enforcement

INV-1 Determinism: Identical inputs produce identical outputs.

INV-2 Identity Primacy: SocioCore's governing rules take precedence over any external instruction.

INV-3 Safety Dominance: Social safety constraints cannot be overridden by obligations, pressure, or preference.

INV-4 Governance Supremacy: Clinical referrals (R8+R7 critical, I4 critical, I10 critical) are not suppressible.

INV-5 Physical Realism: All guidance is achievable within current social state and life context.

INV-6 Domain Integrity: Four primitives are maintained as distinct. No masking of degradation.

INV-7 Abstraction Coherence: Node states derived from available input data via normalized mapping.

INV-8 Semantic Graph Consistency: Cross-node interactions evaluated bidirectionally.

INV-9 Ontology Alignment: All definitions consistent with published SocioCore specification.

INV-10 Safety-First Failure: Missing data defaults to caution-band.

12 Implementation Considerations

12.1 Data Availability Tiers

- **Full integration:** NeuroCore and BioCore coupling active, social event logging, daily structured self-report. Highest precision.
- **Two-layer linked:** NeuroCore active, BioCore absent. Biological coupling nodes default to caution-band.
- **Self-report only:** No organism coupling. All 42 nodes from structured questionnaire. Conservatism multipliers active.

12.2 Clinical Escalation Protocol

Mandatory triggers:
R8+R7 both critical → social isolation crisis referral
I4 critical → identity stability professional referral
I10 critical → urgent psychological evaluation
R8+R2 both critical → clinical evaluation strongly recommended

13 Evaluation Framework

Phase 1: Validate self-report mappings against Social Fatigue Scale, Self-Concept Clarity Scale, UCLA Loneliness Scale, and Perceived Social Support scale.

Phase 2: Validate cross-organism coupling coefficients against published literature.

Phase 3: Deploy with 50–100 practitioners across diverse social contexts over 8–12 weeks.

Phase 4: Track outcomes over 6–12 months. Primary metrics: R8 trend, S8 avoidance, I10 trend, C11 trend.

Appendix A — Complete Node Threshold Table

Node	Description	Advisory	Caution	Critical
S1	Social Battery Level	slightly low	low	depleted
S2	Interaction Fatigue	mild	high	extreme
S3	Social Overload	mild	high	extreme
S4	Social Isolation Pressure	slight	moderate	strong
S5	Connection Warmth	slightly low	low	absent
S6	Presence/Attunement	slightly reduced	poor	absent
S7	Social Recovery Speed	slightly slow	slow	stalled
S8	Social Burnout Risk	possible	likely	active
S9	Social Curiosity	slightly reduced	low	absent
S10	Social Energy Stability	mild variance	unstable	chaotic
I1	Self-Consistency	mild inconsistency	inconsistent	fragmented
I2	Role Stability	mild ambiguity	conflicted	collapsed
I3	Masking Load	mild	high	extreme
I4	Identity Drift	slight	noticeable	severe
I5	Self-Trust	slightly low	low	absent
I6	Boundary Strength	slightly weak	weak	absent
I7	Authenticity Pressure	mild	high	extreme
I8	Identity Recovery Speed	slightly slow	slow	stalled
I9	Narrative Coherence	slightly fragmented	fragmented	incoherent
I10	Identity Safety Index	slightly reduced	low	unsafe
R1	Trust Level	slightly reduced	low	broken
R2	Conflict Pressure	mild	high	extreme
R3	Emotional Contagion	slightly elevated	high	extreme
R4	Relational Predictability	slightly reduced	low	absent

R5	Attachment Security	slightly anxious	insecure	disorganized
R6	Relational Overwhelm	mild	high	extreme
R7	Social Support Strength	slightly low	low	absent
R8	Loneliness Load	mild	moderate	severe
R9	Relational Recovery Speed	slightly slow	slow	stalled
R10	Reciprocity Balance	mild imbalance	significant	extreme
R11	Relational Safety Index	slightly reduced	low	unsafe
C1	Messaging Pressure	mild	high	extreme
C2	Emotional Labor Load	mild	high	extreme
C3	Miscommunication Risk	possible	likely	active
C4	Social Obligations Load	mild	high	extreme
C5	Group Dynamics Stress	mild	high	extreme
C6	Social Role Conflict	mild	significant	severe
C7	Communication Clarity	slightly reduced	poor	absent
C8	Social Bandwidth	slightly reduced	low	exhausted
C9	Conversation Fatigue	mild	high	collapse
C10	Communication Recovery	slightly slow	slow	stalled
C11	Global Social Resilience	slightly low	low	very low

Appendix B — Normalized State Mapping

```
// Subjective nodes — self-report 1–10
function normalize_subjective(rating):
    if rating ≥ 8:  return lerp(+0.3, +1.0, (rating − 8) / 2)
    if rating ≥ 6:  return lerp(0.0, −0.3, (6 − rating) / 2)
    if rating ≥ 3:  return lerp(−0.4, −0.7, (6 − rating) / 3)
    else:            return lerp(−0.8, −1.0, (3 − rating) / 3)
// Event-derived nodes (R2, C1, C4, C5)
function normalize_event_derived(event_log, window_days,
impact_weights):
    weighted_sum = sum(event.impact × impact_weights[event.type]
                       for event in event_log if event.age_days ≤
window_days)
    return map_to_normalized_scale(weighted_sum, threshold_table)
// Coupling nodes
function normalize_coupled(source_state, coupling_coefficient):
    return source_state × coupling_coefficient
```

Appendix C — Social Engine Decision Logic

```
function generate_social_guidance(SC, R, trend, neurocore_state,
biocore_state):
    // Step 1: Organism coupling intake
    R.S2 = weighted_combine(R.S2_selfreport, biocore_state.M8,
biocore_state.S2)
    R.C2 = weighted_combine(R.C2_selfreport, neurocore_state.L2)
    R.I4 = weighted_combine(R.I4_selfreport, neurocore_state.E3)
    R.R2 = weighted_combine(R.R2_selfreport, neurocore_state.E5,
neurocore_state.E6)
    // Step 2: Recompute primitives
    SC = compute_primitives(R)
    // Step 3: Critical pattern detection
    fragmentation_risk = (R.I4 ≤ −0.4 and R.R2 ≤ −0.4 and R.C2 ≤
−0.4)
    burnout_risk = (R.S3 ≤ −0.4 and R.C1 ≤ −0.4 and R.R6 ≤ −0.4)
    // Step 4: Hard constraints
    critical_nodes = [n for n in R if n.state ≤ −0.8]
    suppress_expansion = bool(critical_nodes) or fragmentation_risk
or burnout_risk
    // Step 5: Mode selection
    mode = select_mode(R, SC, fragmentation_risk, burnout_risk)
    // Step 6–8: Generate, filter, score guidance
    candidates = get_mode_guidance_pool(mode)
    if suppress_expansion:
        candidates = filter(candidates, lambda g: not
g.expands_social_engagement)
    for g in candidates:
        g.score = (alpha × g.equilibrium_support − beta ×
g.social_load_added
                   + gamma × trend_alignment(g, trend) − delta ×
g.identity_risk)
    return SocialOutput(mode, sorted(candidates)[:5],
drift_warnings(R, trend))
```

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Patent Pending — Patent Pending.

Patent Pending — DarkWave Studios LLC

Lume-X: Provisionally patented deterministic control runtime.

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